normal subgroup of A. It is clear that $B = K/H$ is a subgroup of $C = L/H$ because K is a subgroup of L.

b. Theorem 34.7 shows that

$$(A/B)/(C/B) \simeq A/C = (G/H)/(L/H) \simeq G/L.$$

9. By Lemma 34.4, we know that $K \vee L = KL = LK$, so $G = KL = LK$. By Theorem 34.5, $G/L = KL/L \simeq K/(K \cap L) = K/\{e\} \simeq K$. Similarly, $G/K = LK/K \simeq L/(L \cap K) = L/\{e\} \simeq L$.

35. Series of Groups

1. (See the answer in the text.)

2. We have to insert subgroups to produce additional factor groups of orders 5 and 49 in the first series, and we create the refinement

$$\{0\} < 14700\mathbb{Z} < 300\mathbb{Z} < 60\mathbb{Z} < 20\mathbb{Z} < \mathbb{Z}$$

of the series $\{0\} < 60\mathbb{Z} < 20\mathbb{Z} < \mathbb{Z}$. We have to insert subgroups to produce additional factor groups of orders 3 and 20 in the second series, and we create the refinement

$$\{0\} < 14700\mathbb{Z} < 4900\mathbb{Z} < 245\mathbb{Z} < 49\mathbb{Z} < \mathbb{Z}$$

of the series $\{0\} < 245\mathbb{Z} < 49\mathbb{Z} < \mathbb{Z}$. These two refinements are isomorphic, producing cyclic factor groups of orders 3, 20, 5, 49, and an infinite cyclic factor group.

3. The given series are already isomorphic, with factor groups of orders 3 and 8.

4. The first series has cyclic factor groups of orders 4, 6, and 3 while the second has cyclic factor groups of orders 3, 2, and 12. Thus we break the 4 into two 2's and break the 12 into a 6 and a 2. We obtain the refinement

$$\{0\} < \langle 36\rangle < \langle 18\rangle < \langle 3\rangle < \mathbb{Z}_{72}$$

of the series $\{0\} < \langle 18\rangle < \langle 3\rangle < \mathbb{Z}_{72}$, and the refinement

$$\{0\} < \langle 24\rangle < \langle 12\rangle < \langle 6\rangle < \mathbb{Z}_{72}$$

of the series $\{0\} < \langle 24\rangle < \langle 12\rangle < \mathbb{Z}_{72}$.

5. (See the answer in the text.)

6. $\{0\} < \langle 30\rangle < \langle 15\rangle < \langle 5\rangle < \mathbb{Z}_{60}$, $\{0\} < \langle 30\rangle < \langle 15\rangle < \langle 3\rangle < \mathbb{Z}_{60}$,
$\{0\} < \langle 30\rangle < \langle 10\rangle < \langle 5\rangle < \mathbb{Z}_{60}$, $\{0\} < \langle 30\rangle < \langle 10\rangle < \langle 2\rangle < \mathbb{Z}_{60}$,
$\{0\} < \langle 30\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{60}$, $\{0\} < \langle 30\rangle < \langle 6\rangle < \langle 2\rangle < \mathbb{Z}_{60}$,
$\{0\} < \langle 20\rangle < \langle 10\rangle < \langle 5\rangle < \mathbb{Z}_{60}$, $\{0\} < \langle 20\rangle < \langle 10\rangle < \langle 2\rangle < \mathbb{Z}_{60}$,
$\{0\} < \langle 20\rangle < \langle 4\rangle < \langle 2\rangle < \mathbb{Z}_{60}$, $\{0\} < \langle 12\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{60}$,
$\{0\} < \langle 12\rangle < \langle 6\rangle < \langle 2\rangle < \mathbb{Z}_{60}$, $\{0\} < \langle 12\rangle < \langle 4\rangle < \langle 2\rangle < \mathbb{Z}_{60}$.

For each series, the factor groups are isomorphic to $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3$, and $\mathbb{Z}_5$ in some order.

7. (See the text answer for the series.) For each series, the factor groups are isomorphic to $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_2$, and $\mathbb{Z}_3$ in some order.

8. There are six possible series $\{(0,0)\} < \langle(m,n)\rangle < \mathbb{Z}_5 \times \mathbb{Z}_5$ where (m,n) is either (0, 1), (1, 0), (1, 1), (1, 2), (1, 3), or (1, 4). There are two factor groups isomorphic to $\mathbb{Z}_5$.

9. (See the text answer for the series.) The factor groups are isomorphic to $\mathbb{Z}_3, \mathbb{Z}_2$, and $\mathbb{Z}_2$ in some order.

10. $\{(0,0,0)\} < \mathbb{Z}_2 \times \{0\} \times \{0\} < \mathbb{Z}_2 \times \mathbb{Z}_5 \times \{0\} < \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$,
$\{(0,0,0)\} < \mathbb{Z}_2 \times \{0\} \times \{0\} < \mathbb{Z}_2 \times \{0\} \times \mathbb{Z}_7 < \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$,
$\{(0,0,0)\} < \{0\} \times \mathbb{Z}_5 \times \{0\} < \{0\} \times \mathbb{Z}_5 \times \mathbb{Z}_7 < \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$,
$\{(0,0,0)\} < \{0\} \times \mathbb{Z}_5 \times \{0\} < \mathbb{Z}_2 \times \mathbb{Z}_5 \times \{0\} < \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$,
$\{(0,0,0)\} < \{0\} \times \{0\} \times \mathbb{Z}_7 < \{0\} \times \mathbb{Z}_5 \times \mathbb{Z}_7 < \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$,
$\{(0,0,0)\} < \{0\} \times \{0\} \times \mathbb{Z}_7 < \mathbb{Z}_2 \times \{0\} \times \mathbb{Z}_7 < \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$.

The factor groups are isomorphic to $\mathbb{Z}_2, \mathbb{Z}_5$, and $\mathbb{Z}_7$ in some order.

11. $\{\rho_0\} \times \mathbb{Z}_2$

12. $\{\rho_0\} \times \{\rho_0, \rho_2\}$

13. $\{\rho_0\} \times \mathbb{Z}_4 \leq \{\rho_0\} \times \mathbb{Z}_4 \leq \{\rho_0\} \times \mathbb{Z}_4 \leq \cdots$

14. $\{\rho_0\} \times \{\rho_0, \rho_2\} \leq \{\rho_0\} \times D_4 \leq \{\rho_0\} \times D_4 \leq \cdots$

15. The definition is correct.

16. The definition is incorrect. It is the factor groups of the series, not the series groups themselves, that need to be abelian.

A **solvable group** G is a group that has a composition series $\{e\} = H_0 < H_1 < \cdots < H_n = G$ such that the quotient groups H_{i+1}/H_i are abelian for $i = 1, 2, \cdots, n-1$.

17. T F T F F T F F T T

18. $\{\rho_0\} \times \{\rho_0\} \leq A_3 \times \{\rho_0\} \leq S_3 \times \{\rho_0\} \leq S_3 \times A_3 \leq S_3 \times S_3$ is a composition series. Yes, $S_3 \times S_3$ is solvable because all the factor groups in this series are of order either 2 or 3 and hence are abelian.

19. Yes, D_4 is solvable, for $\{\rho_0\} \leq \{\rho_0, \rho_2\} \leq \{\rho_0, \rho_1, \rho_2, \rho_3\} \leq D_4$ is a composition series with all factor groups of order 2 and hence abelian.

20. *Chain* (3)
$\{0\} \leq \{0\} \leq \langle 12\rangle$
$\leq \langle 6\rangle \leq \langle 3\rangle$
$\leq \langle 3\rangle \leq \mathbb{Z}_{36}$

Chain (4)
$\{0\} \leq \{0\} \leq \langle 18\rangle \leq \langle 18\rangle$
$\leq \langle 6\rangle \leq \langle 3\rangle \leq \mathbb{Z}_{36}$

Isomorphisms:
$\{0\}/\{0\} \simeq \{0\}/\{0\} \simeq \{0\}$, $\langle 12\rangle/\{0\} \simeq \langle 6\rangle/\langle 18\rangle \simeq \mathbb{Z}_3$,
$\langle 6\rangle/\langle 12\rangle \simeq \langle 18\rangle/\{0\} \simeq \mathbb{Z}_2$, $\langle 3\rangle/\langle 6\rangle \simeq \langle 3\rangle/\langle 6\rangle \simeq \mathbb{Z}_2$,
$\langle 3\rangle/\langle 3\rangle \simeq \langle 18\rangle/\langle 18\rangle \simeq \{0\}$, $\mathbb{Z}_{36}/\langle 3\rangle \simeq \mathbb{Z}_{36}\ /\langle 3\rangle \simeq \mathbb{Z}_3$

21. *Chain* (3)
$\{0\} \leq \langle 12\rangle \leq \langle 12\rangle \leq \langle 12\rangle$
$\leq \langle 12\rangle \leq \langle 12\rangle \leq \langle 4\rangle$
$\leq \langle 2\rangle \leq \mathbb{Z}_{24} \leq \mathbb{Z}_{24}$

Chain (4)
$\{0\} \leq \langle 12\rangle \leq \langle 12\rangle \leq \langle 6\rangle$
$\leq \langle 6\rangle \leq \langle 6\rangle \leq \langle 3\rangle$
$\leq \langle 3\rangle \leq \mathbb{Z}_{24} \leq \mathbb{Z}_{24}$

Isomorphisms:
$\langle 12\rangle/\{0\} \simeq \langle 12\rangle/\{0\} \simeq \mathbb{Z}_2$, $\langle 12\rangle/\langle 12\rangle \simeq /\langle 6\rangle/\langle 6\rangle \simeq \{0\}$,

$\langle 12\rangle/\langle 12\rangle \simeq \langle 3\rangle/\langle 3\rangle \simeq \{0\}$, $\langle 12\rangle/\langle 12\rangle \simeq \langle 12\rangle/\langle 12\rangle \simeq \{0\}$,
$\langle 12\rangle/\langle 12\rangle \simeq \langle 6\rangle/\langle 6\rangle \simeq \{0\}$, $\langle 4\rangle/\langle 12\rangle \simeq \mathbb{Z}_{24}/\langle 3\rangle \simeq \mathbb{Z}_3$,
$\langle 2\rangle/\langle 4\rangle \simeq \langle 6\rangle/\langle 12\rangle \simeq \mathbb{Z}_2$, $\mathbb{Z}_{24}/\langle 2\rangle \simeq \langle 3\rangle/\langle 6\rangle \simeq \mathbb{Z}_2$,
$\mathbb{Z}_{24}/\mathbb{Z}_{24} \simeq \mathbb{Z}_{24}/\mathbb{Z}_{24} \simeq \{0\}$

22. Let $a \in H^* \cap K$ and let $b \in H \cap K$. Then $b \in H$ and $a \in H^*$ so $bab^{-1} \in H^*$ because H^* is a normal in H. Also $b \in K$ and $a \in K$ so $bab^{-1} \in K$. Thus $bab^{-1} \in H^* \cap K$, so $H^* \cap K$ is a normal subgroup of $H \cap K$.

23. We use induction to show that $|H_i| = s_1 s_2 \cdots s_i$ for $i = 1, 2, \cdots n$. For $n = 1, s_1 = |H_1/H_0| = |H_1/\{e\}| = |H_1|$ because each coset in $H_1/\{e\}$ has only one element. Now suppose that $|H_k| = s_1 s_2 \cdots s_k$ for $k < i \leq n$. Now H_i/H_{i-1} consists of s_i cosets of H_{i-1}, each having $|H_{i-1}|$ elements. By our induction assumption, $|H_{i-1}| = s_1 s_2 \cdots s_{i-1}$. Thus $|H_i| = |H_{i-1}|s_i = s_1 s_2 \cdots s_{i-1} s_i$. Our induction is complete and the desired assertion follows by taking $i = n$.

24. By Definition 35.1, a composition series for G contains a *finite* number of subgroups of G. If G is infinite and abelian, and $\{e\} = H_0 < H_1 < H_2 < \cdots < H_n = G$ is a subnormal series, the factor groups H_i/H_{i-1} cannot all be of finite order for $i = 1, 2, \cdots, n$, or $|G|$ would be finite by Exercise 23. Suppose $|H_k/H_{k-1}|$ is infinite. Now every infinite abelian group has a proper nontrivial subgroup and hence a normal subgroup. To see this we need only consider the cyclic subgroup $\langle a\rangle$ for some $a \neq e$ in the group. If $\langle a\rangle$ is finite, we are done. If $\langle a\rangle$ is infinite cyclic, then $\langle a^2\rangle$ is a proper subgroup. Thus we see that H_k/H_{k-1}, as an infinite abelian group, has a proper nontrivial subgroup, so it is not simple and our series is not a composition series.

25. Let $G = G_1 \times G_2 \times \cdots \times G_m$ and suppose that G_i is solvable for $i = 1, 2, \cdots, m$. We form a composition series for G as follows: Start with $H_0 = \{e_1\} \times \{e_2\} \times \cdots \times \{e_m\}$ where e_i is the identity of G_i. Let $H_1 = H_{11} \times \{e_2\} \times \cdots \times \{e_m\}$ where H_{11} is the smallest nontrivial subgroup of G_1 in a composition series $\{e_1\} < H_{11} < H_{12} < \cdots < H_{1n_1} = G_1$ for G_1. Continue to build the composition series for G by putting these subgroups H_{1i} in sequence in the first factor of the direct product series until you arrive at $G_1 \times \{e_2\} \times \cdots \times \{e_n\}$. Then start putting the sequence of subgroups $H_{21}, H_{22}, \cdots, H_{2n_2}$ in a composition series for G_2 into the second factor until you arrive at $G_1 \times G_2 \times \{e_3\} \times \cdots \times \{e_m\}$. Continue in this way across the factors in the direct product until you arrive at

$$G = H_{1n_1} \times H_{2n_2} \times \cdots \times H_{mn_m} = G_1 \times G_2 \times \cdots \times G_m.$$

A factor group formed from two consecutive terms of this series for G is naturally isomorphic to one of the factor groups in a composition series for one of the groups G_i by our construction. Thus these factor groups are all simple so we have indeed constructed a composition series for G. Because all the factor groups in the composition series for G_i are abelian for $i = 1, 2, \cdots, m$ we see that the factor groups of the composition series for G are abelian, so G is a solvable group.

26. Following the hint, Exercise 22 shows that $K \cap H_i$ is a normal subgroup of $K \cap H_{i+1}$ for $i = 0, 1, \cdots, n-1$, so the subgroups $K \cap H_{i+1}$ form a subnormal series for K. Taking $N = H_{i-1}$ and $H = K \cap H_i$ as subgroups of H_i, and applying Theorem 34.5, we see that $HN/N = [(K \cap H_i)H_{i-1}]/H_{i-1} \simeq H/(H \cap N) = (K \cap H_i)/(K \cap H_{i-1})$. Now $(K \cap H_i)H_{i-1} \leq (K \cap H_i)H_i = H_i$ so $[(K \cap H_i)]/H_{i-1}$ can be viewed as a subgroup of H_i/H_{i-1}. Because H_i/H_{i-1} is a simple abelian group, we see that $[(K \cap H_i)H_{i-1}]/H_{i-1}$ is either the trivial group or is isomorphic to H_i/H_{i-1}, and hence is simple and abelian because G is solvable. Thus the distinct groups among the $K \cap H_i$ form a composition series for K with abelian factor groups, and consequently K is solvable.